

AI/ML Internship Report – Virtual Try-On System

This report explains the architecture and workflow of an AI-powered Virtual Try-On system developed using the DeepFashion2 dataset. The project focuses on creating a pipeline capable of detecting body pose, segmenting garments, warping clothing according to body structure, and generating realistic final try-on images. The report is designed according to internship sprint documentation standards and includes technology stack, module responsibilities, APIs, and intern roles.

Project Objective

The objective of the project is to build a working virtual try-on AI pipeline that can:

- Remove existing clothing from a person image
- Generate cloth-agnostic person representations
- Detect body pose and landmarks
- Warp garments according to body posture
- Blend garments naturally onto the person image
- Produce realistic final try-on outputs

Complete AI Pipeline

Person Image → Pose Detection → Body Parsing → Cloth Segmentation → Agnostic Mask Generation → TPS Garment Warping → Try-On Blending → Final Output Image

Module 1 – Dataset Preparation & Data Processing

The first notebook focuses on preparing the DeepFashion2 dataset for training and testing. This stage ensures that all images, masks, and annotations are properly structured before being passed into AI models. **Main Tasks:**

- Loading image datasets and JSON annotations
- Creating structured DataFrames using Pandas
- Cleaning corrupted or missing records
- Image resizing and normalization
- Organizing train, validation, and test datasets

Tech Stack:

Python, Pandas, NumPy, OpenCV, DeepFashion2 Dataset

Role Relevance:

Data Interns are responsible for dataset management, annotation handling, preprocessing, and ensuring the AI pipeline receives clean and structured data.

Module 2 – Human Parsing & Cloth Segmentation

The second notebook focuses on identifying body regions and separating clothing from the background. This stage is essential because the system must understand where the human body and garments exist before applying a new outfit. **Main Tasks:**

- Human body detection
- Pose estimation using keypoints
- Cloth segmentation and mask generation
- Upper and lower garment separation
- Agnostic person image generation

Pose Detection Keypoints:

Shoulders, elbows, hips, knees, ankles, waist positions

Tech Stack:

MediaPipe, OpenCV, U-2-Net, Segment Anything Model (SAM), Pillow

Role Relevance:

Computer Vision Interns work on segmentation quality, pose detection accuracy, mask postprocessing, and body parsing optimization.

Module 3 – Garment Warping Engine

This module is the core component of the virtual try-on system. The AI model reshapes the selected garment according to body posture and perspective. **Main Tasks:**

- TPS transformation implementation
- Warp control point generation
- Geometric matching between garment and body
- Pose-based cloth alignment
- Edge case handling for difficult poses

Warping Techniques:

Thin Plate Spline (TPS), Geometric Matching Module (GMM), Flow-based warping

Tech Stack:

PyTorch, HR-VITON, VITON-HD, TPS Transform, DressCode Dataset

Role Relevance:

ML Interns are responsible for implementing TPS warping, optimizing geometric alignment, testing edge cases, and validating warping outputs.

Module 4 – Try-On Blending & Composition

After warping the garment, the final stage blends the clothing naturally onto the person image. This stage improves visual realism and ensures smooth garment integration. **Main Tasks:**

- Alpha blending and image composition
- Preserve face, hair, and body regions
- Texture consistency and shadow generation
- Final image enhancement and rendering

Alpha Blending Formula:

$$I = \alpha F + (1 - \alpha)B$$

F = Warped Garment

B = Background/Person Image

Tech Stack:

HR-VITON, ACGPN, OpenCV Alpha Blending, Pillow

Role Relevance:

AI Leads and Integration Interns handle pipeline integration, API communication, dependency management, debugging, and complete workflow validation.

API Integration Workflow

The complete system is connected using backend APIs. **Segmentation API:**

POST /ai/segment-cloth

Warping API:

POST /ai/warp-garment

Final Try-On API:

POST /ai/tryon

These APIs allow communication between segmentation modules, warping engines, and final try-on generation systems.

Technology Stack

Programming Language: Python

AI & Deep Learning:

- PyTorch
- GANs
- U-Net
- HR-VITON
- VITON-HD

Computer Vision:

- OpenCV
- MediaPipe
- Segment Anything Model (SAM)
- U-2-Net

Data Handling:

- Pandas
- NumPy

Development Platforms:

- Kaggle
- Google Colab
- GitHub

Backend Integration:

- Flask APIs
- FastAPI
- JSON APIs

Final Deliverables

- Working Virtual Try-On Pipeline
- Cloth Segmentation System
- Pose Estimation Module
- TPS Garment Warping Engine
- Final Try-On Output Generator
- API Integration Workflow
- End-to-End Testing Pipeline

Conclusion

The Virtual Try-On system combines Computer Vision, Deep Learning, Pose Estimation, Image Segmentation, and AI image generation into a complete AI workflow. The project directly aligns with internship responsibilities such as dataset preparation, AI model testing, pipeline integration, API development, segmentation research, workflow debugging, and end-to-end validation. The project demonstrates how modern AI systems can transform the fashion and e-commerce industry by creating realistic digital fitting room experiences.

Role	Responsibilities
AI Lead	Architecture, APIs, integration, debugging, final demo
Computer Vision Intern	Segmentation, masking, body parsing
ML Intern	TPS warping engine, geometric matching
Data Intern	Dataset preparation, testing, annotations